Ninel **BĂNICĂ**

Embedded & Software Engineer

CONTACT & INFO

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/Ninel-Valentin

ninel-valentin.com

LANGUAGES

Romanian	**Fluent**
English	**C1**
French	**B1**

SOFT SKILLS

- Problem-Solving
- Quick learner
- Curiosity
- Communication
- Time Management

TECHNICAL SKILLS

	Beginner - Expert
PCB/Circuit design	●●●
Measurements	●●●●
Power electronics	●●●
Digital circuits	●●●
Analog circuits	●●●
Verilog	●●●
Vivado/Vitis	●●
C/C++	●●●●
Debugging	●●●●●

Many more projects that I've worked on since 2021 available on my portfolio website.

Education

Bachelor of
Electrical Engineering & Computer Science
Transylvania University of Brasov **2022 - 2026**

Experience

FPGA development
Univeristy Transylvania of Brasov **Jun 2025 - Sep 2025**

- Programmed FPGA using Vitis and Vivado;
- Learnt about SEU effects on hardware, error correction, error pattern observation and prevention;
- Observed and learnt from an EU project research of a custom FPGA design.

Intern
Siemens Digital Industries Software

Data science & Research **Jun 2025 - Present**

- Researching newly acquisitioned tools from Altair;
- Researching optimization methods for predictive models for power consumption of full electric/hybrid sea vessels.

Digital twin Developer **Jun 2025 - Present**

- Improved a digital twin excavator arm's electrical configuration;
- Integrated an external ADC chip on a custom RaspberryPi Hat PCB.
- Improved the code and scripts as for the arm and AmeSim simulation to work on the RaspberryPi without an external computer;

Fullstack Developer **Jun 2025 - Present**

- Created few additional add-on modules/windows for Testlab, a C# and C++ software for data analysis of automotive sensors;

Summer school **Jun 2025 - Present**

- Developed multiple microservices using Python and deployed them online using AWS;

Junior Crawler Developer
ChannelSight **Mar 2022 - Nov 2024**

- Developed and improved Web-Crawlers on Apify using NodeJs.
- Wrote custom JavaScript tools to ensure data quality in large volumes.
- Created SQL queries for data preview.

Certifications/Courses

- AWS Cloud Practitioner **Oct 2024**
- SQL Server fundamentals - PluralSight **Jul 2023**

Personal projects

Embedded system for satellite imagery and sensor data transmission 2025

- Designed a multi-sensor, dual-cam (*video and IR*), ESP32 controlled, PV charging battery system PCB transmitter with a trace antenna;
- Designed an ESP32 controlled receiver with a integrated trace antenna for downloading the satellite data to a laptop;
- Simulated and tunned RF transmission line in Ansys Electrical.
- Programmed data processing and transmission protocols.

Gyrosensor controlled AVC recovering device 2025

- Built a battery powered gyrosensor glove which sends control commands via bluetooth to a separate LCD application.
- The application features few games to train dexterity;

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